

UAS Air-Risk Assessment In and Around Airports

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Abstract—The operation of Uncrewed Aircraft Systems (UAS) at airports is becoming more common. The increased use of small and inexpensive drones pose various challenges associated with integrating UAS operations in the national airspace system. For non-military applications, Uncrewed Aircraft (UA)s are allowed to fly below 400ft which helps to segregate UAS operations from crewed aircraft. This however does not negate the challenges and additional risk in low altitude scenarios such as in and around airports when UASs are used for commercial and security applications. Flying the UASs in such controlled airspace needs permission from authorities such as Air Traffic Control (ATC) and other controlling agencies. For such scenarios where UAS operations need to be integrated along with controlled airspace, it is important to understand and estimate the associated risk. An unintentional malfunction resulting in uncontrolled UASs poses multiple risks, particularly when operated in a busy airport environment. This includes infrastructure, ground, and air risk. When left unmitigated, such scenarios will lead to the disruption of regular operations and cause loss to the economy and sometimes human life. The paper focuses on developing an assessment tool for UAS collision risk with crewed aircraft in an airport scenario. The study focuses on the UAS risk associated with crewed aircraft flying below 1000 ft altitude within a 5 mile radius of an airport. High volume airport operations combined with low-altitude flights results in an increased risk of collision within an airport environment. The trajectory of aircraft in three-dimensional space is determined using actual historical Automatic Dependent Surveillance-Broadcast (ADS-B) data. Simulations are conducted to model various fail-safe scenarios of UASs. A probabilistic approach is used to model UA paths that assign Gaussian distributions to the mean values of the UAS's velocity, heading, and altitude. The framework developed can be further expanded to include specific waypoints or routes other than ADS-B data. The study aims to calculate the probabilities of Mid-Air Collisions (MAC), Near Mid-Air Collisions (NMAC), and Well Clear (WC) violations entering these protected volumes between uncrewed and crewed aircraft. Initially, the study modeled the Grand Forks International Airport, which has a high volume of airport operations. Historical ADS-B data is analyzed statistically to identify the highest volume of traffic for a given day. Flight trajectories from that time interval are then extracted for analysis. The UAS-flight risk assessments are carried out for various scenarios that include the velocity of the aircraft, traffic volume, and the probability distributions of the UAS's trajectory.

Index Terms—Unmanned Aircraft, Risk assessment, Airport operations

I. INTRODUCTION

Uncrewed Aircraft Systems (UAS), are becoming increasingly popular for various applications. However, operating UAS near airports can pose significant risks to crewed aircraft, passengers, and property on the ground. To ensure safe and

efficient UAS operations near airports, there are guidelines and regulations that must be followed [1]. Federal agencies such as Federal Aviation Administration (FAA) in the United States have established guidelines for operating UA in or near the airport [2]. This includes obtaining prior permission from FAA and ATC before flying within the airport. The basic rule while flying UA at airports includes that priority is given to the crewed aircraft while maintaining a visual line-of-sight with the UA at all times. While operating a UA within the airport, UAS operators must also ensure that their drones are equipped with technology such as geo-fencing, which prevents the drone from entering the volume of airspace which is beyond its operational limit [3]. In addition, the operators are highly recommended to monitor weather conditions such as wind and have a plan in place for emergency landings. Operators are also recommended to be in reliable communication with ATC while operating. This assists ATC to be aware of any emergency scenario such as a lost link or GPS failure or similar. Failure to do so could result in a life-threatening scenario such as a MAC with a crewed aircraft.

The following incidents near the airports indicate how UAS disrupts airport operations. In December 2018, London's Gatwick Airport was closed for a few hours due to drone sightings near the runway, causing more than thousands of flights to be canceled or diverted [4]. In January 2019, Newark Liberty International Airport in New Jersey was temporarily closed after two pilots reported seeing a drone flying at an altitude of about 3,500 feet. The airport was closed for about an hour while authorities searched for the drone and its operator [5]. Early 2020, flights were forced to make an emergency landing at Madrid's Barajas Airport after a drone was noticed near the runway which resulted in delaying the operation and re-routing flights to other airports [6]. In February 2020, Dubai International Airport was briefly closed due to unauthorized drone activity in the vicinity. In March 2022 all flight operations at Dublin Airport were stopped because of a drone sighting in the airport airspace [7]. The airport was closed for about 20 minutes, causing several flights to be delayed. These incidents depict how an unexpected drone impacts an airport operation and can create a huge risk to human life and also results in heavy economic loss.

Though drone usage in and near airport poses a significant risk to crewed aircraft, they are valuable tools for a variety of applications including surveillance, inspection, and more. Though some government agencies issued no-fly zone for drones, FAA allows UAS operations with prior permissions. More research needs to be done to improve the safe operations

of drones in airports by studying the risk analysis based on the altitude, distance, time of day, aircraft traffic patterns, etc. The airport applications include inspection and maintenance where UAS equipped with sensors such as cameras, LIDAR, etc can be used to inspect airport infrastructure such as runways, taxiways, and buildings and hence to reduce the risk of accidents and thus ensure the smooth operation of the airport [8]. UAS platforms are also used for security and surveillance purposes such as monitoring the boundary of the airport to detect intruders and other potential security threats [9] [10]. Wildlife management is another potential application for controlling wildlife including birds in and around the airport. UAS has been used to support airfield operations such as providing real-time conditions such as weather, runway, and aircraft movement. Cargo delivery and passenger transportation are other major application areas where UAs could be used in airport environments [11]. The use of drones near airports is a highly sensitive issue, and all entities involved must coordinate together to ensure that the use of UAs does not compromise the safety and security of airspace.

A failure in operation such as a lost link or GPS failure can result in a loss of a positive aircraft control which may result in entering the operational volume of crewed aircraft. One of the main threats of a lost drone can be a collision with crewed aircraft or with critical infrastructure. Collision with a crewed aircraft can bring an aircraft down causing a potential threat to the human-life, ground infrastructure, and other casualties. Based on the categories, the risk can be mainly classified risk into Air risk, Infrastructure Risk, and Ground risk [12] [13]. The paper presents the collision risk analysis caused by a non-cooperative UAS with a crewed aircraft during operational hours.

The remaining sections are organized as follows. Section II discusses the background. The methodology is explained in section III where the different metrics used for the analysis were discussed. Section IV discusses the simulation scenarios and the results and finally, section V concludes the paper.

II. BACKGROUND

The paper considers the use of the probability risk analysis to access the likelihood of an event occurring when an uncrewed aircraft is gone out of control while operating in an airport environment. Aviation risk assessment was done in various studies [14] [15]. The section discusses the similar line of work done by researchers in estimating risk assessment when UA are operated along with other crewed aircraft and also in airport scenarios.

Collision risk assessment between UAS and a landing aircraft was done by [16] using 3D monte-carlo simulation. Collision probability at each time step under different scenarios is considered for the analysis in the paper. A 3D movement model of a single crewed aircraft was fed into the simulation to calculate the probability analysis. The paper [17] discusses the use of ground-based tracking equipment to improve collision prediction accuracy. The paper discusses how the UAS tracking information could be used to complement the collision

prediction algorithm and how its inclusion affects the collision risk assessment.

The paper [18] discusses the important risks related to conducting operations with the use of UAS by first responders (FRs) and also presented the risk analysis for similar operations. The paper considered the possibilities offered by UASs in crisis or disaster management and step-by-step Specific Operations Risk Assessment (SORA) and also the consequences, methods, and mitigation methods during the FR missions. The risk assessment of drone intrusion in airports was done in [19]. The paper discusses the risk assessment of drone intrusion based on drone-intrusion features, airport features, and current operations and by considering both safety and security-related causes. Domenico et al., discuss the quantitative assessment of the historical feature of drone intrusion in airports by using different public databases [20]. The authors use probability distribution to model the available features and calculated risk analysis using supervised machine learning. UAS collision risk assessment in restricted areas of an airport was done by [21]. The authors use the dynamically generated 3D trajectories of the UAS encounter to assess the collision risk. A collision risk heatmap representation is shown to predict the collision risk at airport-restricted areas. Other than air risk, ground risk is one of the key parameters for assessing the risk before the UAS operations. The paper [22] talks about a dynamic model based on a deep learning approach to assess the ground risk. The paper uses a kinetic-theory-based model to assess the ground impact.

The presented work uses the real-world traffic data of Grand Forks International airport. Peak operation hour data is considered for the analysis. The ADS-B data is used to identify the peak traffic and hence to calculate different aircraft trajectories. ADS-B is a surveillance technology that utilizes radio signals to periodically transmit essential information, including GPS location, altitude, speed, and other flight parameters of an aircraft. These attributes enable the aircraft to be tracked effectively. *ADS-B out* refers to the transmission of information from the aircraft, while *ADS-B in* refers to the reception of information from other aircraft.

III. METHODOLOGY

Probabilistic risk assessment (PRA) is used in the present study as a methodical and all-inclusive procedure utilized to assess the hazards related to UAS operations with moving aircraft within the airport environment given a well-clearance violation [23]. Risk within the realm of PRA is defined as an attainable unfavorable consequence of an undertaking or operation. In a PRA, the risk is distinguished by two quantities: i) the magnitude or intensity of the potentially unfavorable outcomes and ii) the likelihood or probability of each outcome occurring. The work presented in the paper focuses on the probability of an event occurring. MAC, NMAC and WC risk volumes associated with the aircraft are identified [23]. All these three risk volumes are represented by concentric cylinders. Dimensions of NMAC and WC volumes are considered from previous studies [23]. MAC volumes are

considered based on the approximate aircraft dimensions in a 3D space. We have also identified different WC risk volumes based on the TCAS and DMOD metrics [24]. But finally, we chose WC as spatial risk volume defined by a radius of 2000 ft and height of 500 ft [23]. Specific details of each of these cylindrical volumes are provided in table I and a schematic 3D and 2D representation is shown in figure 1. The probabilities associated with these three risk volumes are calculated in different simulation settings. Methodologies related to probability calculation, uncrewed aircraft trajectory generation, and aircraft trajectory extraction from ADS-B data are presented in the following sections.

TABLE I
RISK VOLUMES AND THEIR DIMENSIONS CONSIDERED IN THE PRESENT STUDY

S.No	Risk volume	Radius (ft)	Height (ft)
1	Mid-air collision (MAC)	40	20
2	Near mid-air collision (NMAC)	500	200
3	Well-clearance (WC)	2000	500

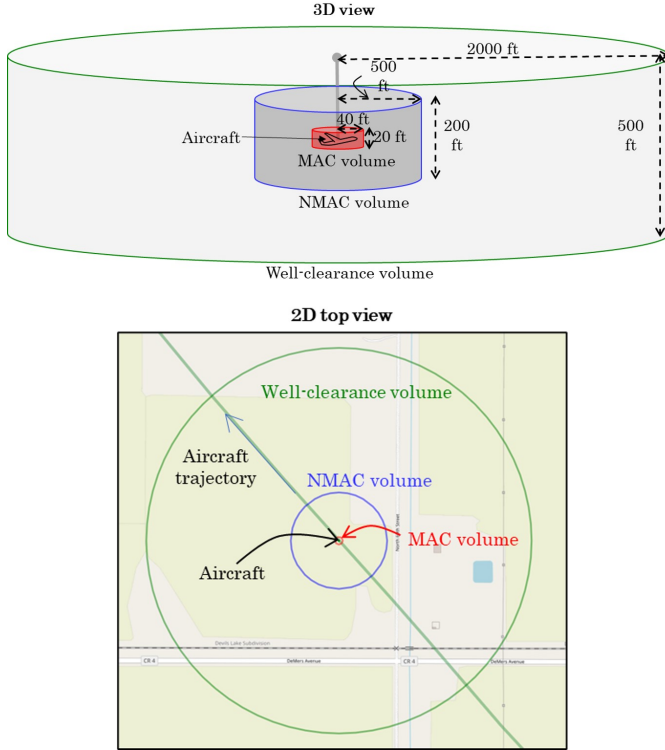


Fig. 1. Schematic diagram showing the different risk volumes and their dimensions used in the present study, 3D view (top) and 2D top view (bottom) (not to scale)

A. Probability calculation

The methodology employed to ascertain the probabilities of each risk volume is as follows: The initial step entails identifying the risk volume corresponding to the aircraft's location at a particular moment. Concurrently, the probable locations of the uncrewed aircraft are generated through the use of a

pre-defined probability distribution function. Subsequently, the likelihood is calculated as the ratio of UAS locations that fall within the risk volume to the total number of UAS locations.

$$P_{(Risk,t)} = \frac{N_{\text{Points in risk volume}}}{N_{TP}} \quad (1)$$

where $P_{(Risk,t)}$ is the probability of UAS within the given risk volume at time t , $N_{\text{Points in risk volume}}$ is the number of random samples located spatially within the given risk volume at time t and N_{TP} is the total number of random samples generated. In this investigation, we have adopted a methodology that bears some similarities to the one used by [23], wherein it is assumed that both the crewed and uncrewed aircraft did not have collision risk prevention systems like transponders or TCAS. However, in this paper, the likelihood of collision is determined as a single measurement, assuming that both the aircraft and the UAS are operating without the knowledge of each other's presence and without any separation maneuver to reduce the risk of collision. To be more precise, each risk probability at a given time t can be expressed in an explicit manner as follows:

$$P(MAC) = \frac{N_{MAC}}{N_{TP}} \quad (2)$$

where N_{MAC} is the number of random samples located spatially within MAC risk volume. and N_{TP} is the total number of points generated using the probability distribution function.

$$P(NMAC) = \frac{N_{NMAC}}{N_{TP}} \quad (3)$$

where N_{NMAC} is the number of random samples located spatially within NMAC risk volume.

$$P(WC) = \frac{N_{WC}}{N_{TP}} \quad (4)$$

where N_{WC} is the number of random samples located spatially within WC risk volume. We have assumed that $P(MAC)$, $P(NMAC)$, and $P(WC)$ are independent of each other. The following sections provide a detailed exposition of the process employed to generate uncrewed aircraft locations at a given point in time, as well as the risk volumes linked to aircraft data.

B. UAS trajectory

A UA is hypothesized to perform an inspection of a building within the airport environment. To simulate the UAS operation, we employed the SIMLAT Software to conduct UV simulations of building inspection [25]. Simulations were conducted to fly UAS in a pre-defined path consisting of waypoints surrounding the building. However, by introducing different failure scenarios, the UA's trajectory was observed to deviate from the original path. These scenarios included GPS failure, battery drain, communication failure between the UA and the pilot, and severe wind conditions. In most of the failures except GPS failure, the UA can be pre-configured to return to safe geolocation. A worst-case scenario can be GPS failure and loss of communication link happening at the same time where

the UA is out of the control of the pilot. A similar scenario can be a wrong configuration given to return to home (RTH) where UV will start to navigate to a un-safe location in case of a failure such as a lost link.

The simulation results revealed that when the UA's GPS failed, the vehicle continued in the same direction it was flying before the failure. In all other scenarios, the UA returned to its home location when a failure occurred. Nonetheless, the UA's precise heading at any given time exhibited slight perturbations from the line connecting the two waypoints. Additionally, the UA was unable to take a perpendicular angle at the corners of the building. Consequently, it is impossible to determine the exact path of the UA following a failure. This naturally prompts the use of a probabilistic approach to assigning a probability to the UAS's failure scenario.

In this investigation, we have examined the instances of UA movement in a completely stochastic direction. Based on the work done by [17] and [16], we have established probability distributions for the UA. Although the work considered both intended and unintended scenarios and employed kinematic equations with parameters specific to UA. The work done by Wang et al., argued that utilizing a random multiplier for the thrust vector in these equations resulted in a normal distribution of UA positions at each time step with a mean value corresponding to the initial velocity. In contrast, we directly produced random normal distributions with mean values determined by the initial velocity and standard deviation. Additionally, our study diverged from [17] and [16] by incorporating an actual flight path extracted from ADSB data.

The velocity and heading of a UAS can be characterized by probability distributions. Specifically, the magnitude of the velocity, denoted as $|v|$, can be expressed as a Gaussian distribution with mean v_{mean} and standard deviation σ_v , according to the following equation:

$$|v| = \frac{1}{\sigma_v \sqrt{2\pi}} \exp \left(-\frac{1}{2} \left(\frac{x - v_{mean}}{\sigma_v} \right)^2 \right) \quad (5)$$

Likewise, the heading direction of the UA, denoted as θ , can be described by another Gaussian distribution with mean θ_{mean} and standard deviation σ_θ , as follows:

$$\theta = \frac{1}{\sigma_\theta \sqrt{2\pi}} \exp \left(-\frac{1}{2} \left(\frac{x - \theta_{mean}}{\sigma_\theta} \right)^2 \right) \quad (6)$$

Similarly, the change in altitude of the UAS is derived using an angle ϕ , which can be described by another Gaussian distribution with mean 0 and standard deviation σ_ϕ , as follows:

$$\phi = \frac{1}{\sigma_\phi \sqrt{2\pi}} \exp \left(-\frac{1}{2} \left(\frac{x}{\sigma_\phi} \right)^2 \right) \quad (7)$$

To generate new positions for the UA, a random number generator is used to sample from these probability distributions. It should be noted that the magnitude of the UA's velocity, heading direction, and altitude change are sampled independently from the respective Gaussian distributions. The

distance traveled by the UA at a particular time is then calculated as the product of velocity and the time

$$d_t = |v| \times t \quad (8)$$

Subsequently the new spatial location of the UA $P_{t=t}(x, y, z)$ in 3D space is calculated using the formula as follows [26]

$$P_{t=t}(x, y, z) = P_{t=0}(x, y, z) + (d_t \sin \theta \cos \phi, d_t \cos \theta \cos \phi, d_t \sin \phi) \quad (9)$$

where, $P_{t=0}(x, y, z)$ is the origin of the UA at time $t = 0(sec)$. A schematic diagram showing the different components of the equation representing the UA trajectory is shown in figure 2.

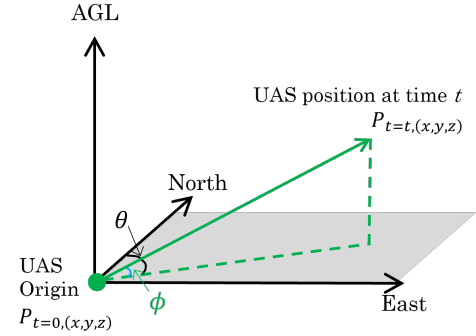


Fig. 2. Graphical representation of an uncrewed aircraft's origin location when the failure occurred and the location after the failure time t .

C. Aircraft traffic

Aircraft traffic is extracted from the ADS-B data. ADS-B data during the months from April to September 2021 is obtained from the ADS-B receiver deployed at the Grand Forks international airport by Flightradar24 [27]. ADS-B data Pre-processing is a major step before the analysis to extract the crewed air traffic in the volume of interest in and near the airport. Data Pre-processing also includes removing the traffic which is stationary, such as on the ground also interpolating the missing data points. For example, the ADS-B received may not be uniformly sampled and needs interpolating the intermediate points to capture a better understanding of when an aircraft takes a turn, ascending or descending into a runway. The choice of sampling interval depends on the specific problem and the desired interval for calculating probabilities. For the work presented in the paper, the ADS-B data has been re-sampled uniformly at every 1-second interval. From the ADS-B data set, the following flight-specific data is extracted: i) time, ii) location (latitude and longitude), and iii) altitude. It is important to note that the altitude above ground level (AGL) is calculated using the ADS-B altitude data using digital elevation maps obtained from USGS EarthExplorer [28].

IV. RESULTS AND DISCUSSION

The result presented in this section uses the statistical analysis of the ADS-B data pertaining to flight traffic over the region within 5 miles of the Grand forks international

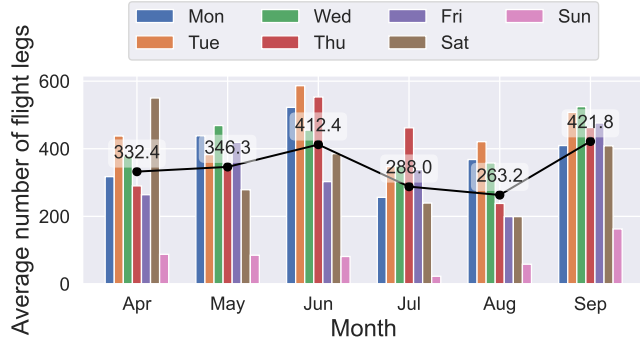


Fig. 3. Monthly statistical histograms corresponding to the Grand Forks International Airport (GFK). Data source: ADSB data obtained from flightradar24.com

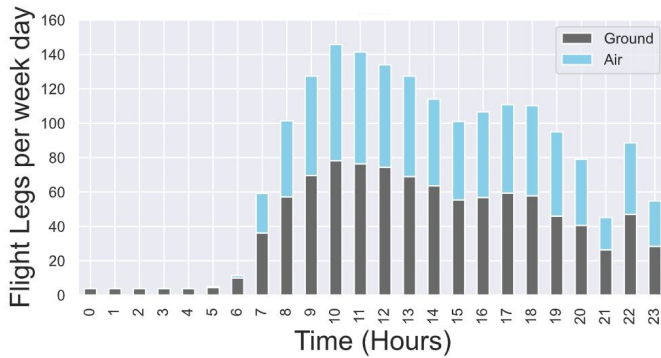


Fig. 4. Hourly statistical histograms corresponding to Tuesdays at the Grand Forks International Airport (GFK). Data source: ADSB data obtained from flightradar24.com

airport (GFK). The analysis is primarily focused on identifying and characterizing the temporal patterns in the air traffic flow. In this regard, the hourly, weekly, and monthly flight traffic data are analyzed. The outcome of the analysis is shown in the figures 3 and 4. The following section presents different scenarios considered for data analysis.

A. Scenario-I: Probabilistic risk assessment - single crewed aircraft and UAS

The objective of this simulation scenario is to present an exposition of the current framework in a straightforward and easily comprehensible manner. The simulation setup consists of a solitary aircraft descending towards the GFK airport, serving as the focus of our risk analysis. The aircraft's flight path is derived from the ADS-B location series as a function of time, following the methodology outlined in the previous section. The trajectory of the aircraft is illustrated in figure 5. As shown in the figure, the aircraft followed a path starting at an altitude near 1000ft at 10:00:26 (CST) and reached a stationary location at 10:09:30. The operation of UAS is simulated along with the real ADS-B traffic to calculate the PRA via an in-house simulation platform developed using

python programming. The simulation platform overlaps the real ADS-B traffic data with the simulation-generated UA trajectories. In the simulation, UA before failure is assumed to have operated from a nearby building within the airport. This is done with the assumption that the UA is used for applications such as building inspection. The simulation parameters used are presented in Table II. In the simulation, the time of UA failure is varied between 0 to 600 seconds.

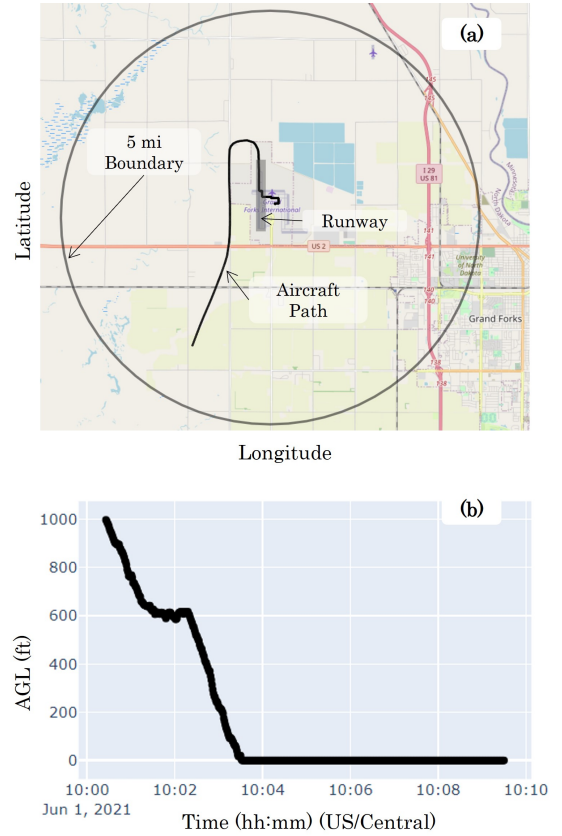


Fig. 5. Aircraft used in the scenario-I (a) Latitude-longitude trajectory of the aircraft (b) Altitude of the aircraft as a function of time

TABLE II
PARAMETERS USED IN THE SIMULATION SCENARIO-I

Paramter	Units	Value
UAS Origin location	Degrees	(47.94962336, -97.17673061)
UAS Elevation	ft	90.0
UAS Fail time	sec	0 to 600 with steps of 10
UAS Velocity mean	ft/sec	20.0
UAS Velocity S.D	ft/sec	2.0
UAS Heading Mean	Degrees	0.0
UAS Heading S.D	Degrees	360.0
UAS Elevation S.D	Degrees	0.5

The probabilities associated with the three risk volumes of the landing path of a single aircraft are then determined. The trajectory of both aircraft are shown in figure 6. The results are presented in figure 7. Figure 7(a) indicates that the $P(MAC)$ is almost zero, while $P(NMAC)$ reaches its peak value of

0.0099, and the value of $P(WC)$ reaches its maximum value of 0.147 when the time = 220s.

It is interesting to note that the probability value is time-dependent, as the UAS is assumed to have failed at a specific time ($t_f = 10:00:00$ CT), and the high-risk periods occur 220 seconds after this failure. The main factors include the crewed traffic in the airport, the traffic pattern of the crewed aircraft, operation location of the UA before the failure occurs. To investigate the effect of the failure time, further simulations were conducted while keeping all the remaining parameters constant and varying the UAS fail time. The location of the aircraft and the UA positions are shown in figure 6. The result presented in figure 7(b) shows the probability value in the simulation with respect to UAS fail time.

From the figure, it is also clear that the $P(MAC)$ is almost zero for different t_f values, and the value of $P(NMAC)$ varies within a small range of 0.0093 to 0.0106. Similarly, the value of $P(WC)$ also varies in a small range of 0.144 to 0.150. These variations are minimal and suggest that the associated risk probabilities remain almost the same irrespective of the UAS failure time during the aircraft landing time.

B. Scenario-II: Probabilistic risk assessment during the peak traffic volume

The flight traffic data from the peak traffic during the summer time period is considered for this scenario. As mentioned in section III, the ADS-B data collected from the Grand Forks international airport is analyzed in order to derive flight traffic statistics. Our observations show that the peak traffic is observed on most Tuesdays during the month of June. As shown in figure 4, the hourly data on Tuesdays show that the highest flight traffic is concentrated within the 10:00 to 11:00 AM CT interval. Subsequently, the flight trajectories that fall within the 10:00:00 to 10:22:00 (CT) time frame are selected for further simulation analysis. The flight trajectories during this location are presented in figure 8. The simulation parameters of the UAS are the same as scenario-I, as shown in the table II. However, in this scenario, only the fail time of 10:00:00 (CT) is investigated.

The probabilities associated with the three risk volumes are calculated when the air traffic is at its peak. The UAS failure time is assumed to be 10:00:00 (CT) and the results are shown in figure 10. Our analysis shows that the $P(MAC)$ value is negligible, the value of $P(NMAC)$ reaches its maximum value of 0.31, while the value of $P(WC)$ is in its maximum value of 1.0. Similar to the previous case, the risk probability is time-dependent, and the highest risk occurs within the first 200 seconds after the UAS failure. The reason for high $P(WC)$ is that the UAS failure location is relatively close to the air traffic as shown in figure 9. The UAS failure location is in close proximity to the center of the airport, where landing and taxing aircraft are concentrated. Hence, the high-risk probabilities occur within the first few minutes of the UAS failure. If the UAS failure location is away from the airport center where more air traffic is present, the lower the traffic density is and hence low probability risk. In the next section, the risk

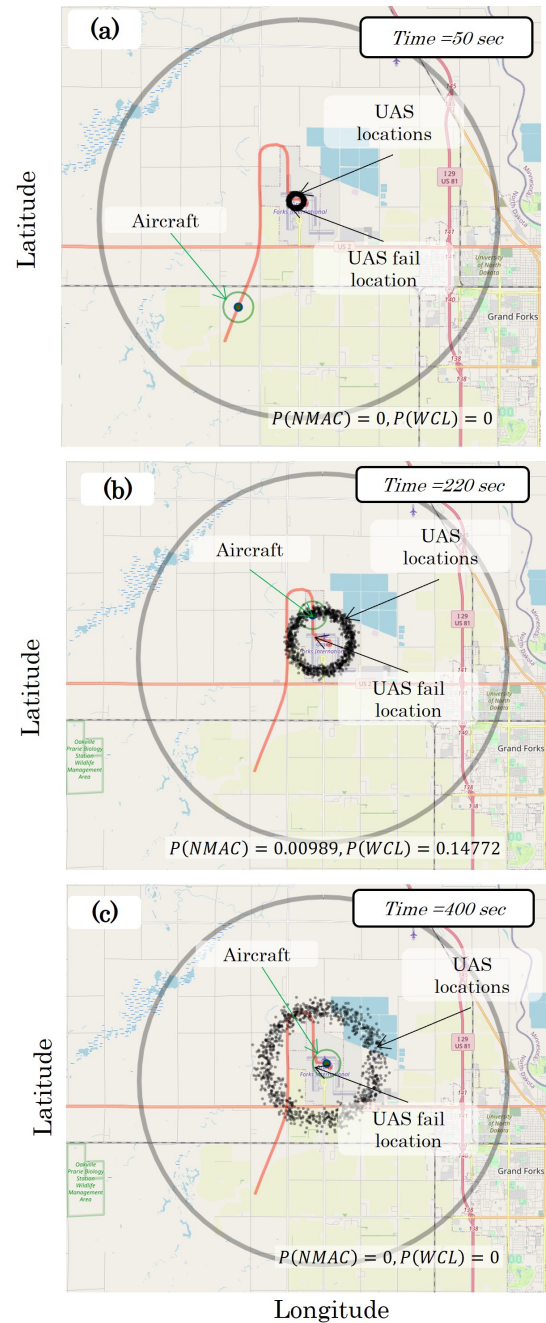


Fig. 6. UA probable locations and aircraft locations at times (a) 50 sec (b) 220 sec and (c) 400 sec.

probabilities associated with UAS failure for the entire airport are calculated by assuming multiple UAS fail locations.

C. Scenario-III: Probabilistic risk assessment - UAS fail location

The primary objective of this simulation scenario is to conduct a probabilistic risk assessment of the UAS based on its operating location. To achieve this goal, we have adopted a structured approach by considering an area of $3mi \times 3mi$ miles around Grand Forks airport. A uniform grid of 20×20 is

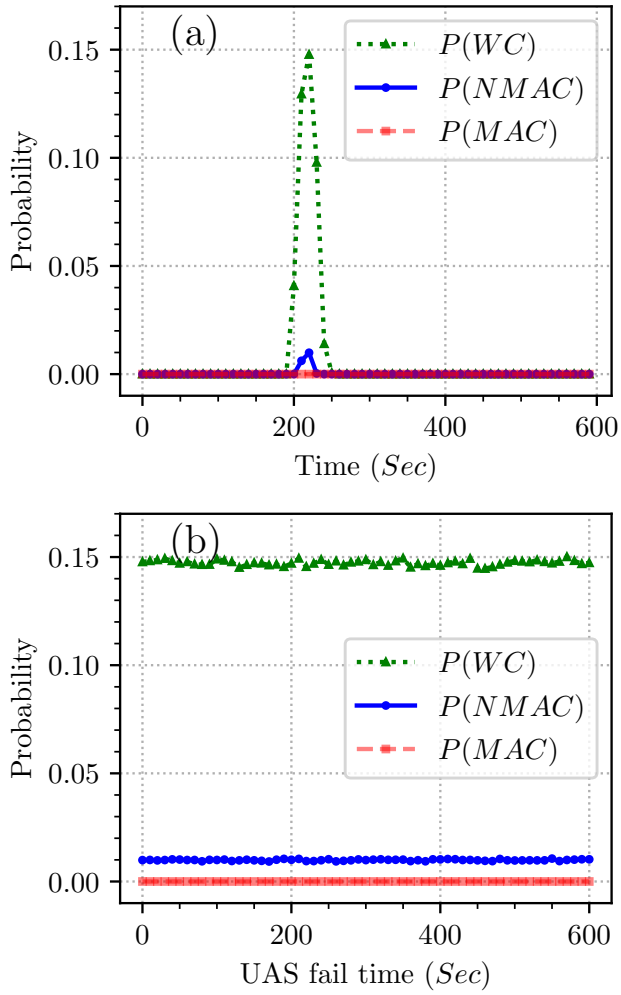


Fig. 7. Probabilities of MAC, NMAC, and WC at when the (a) UA failure occurred at '0' seconds (b) Peak probabilities when UA failure occurred at different times

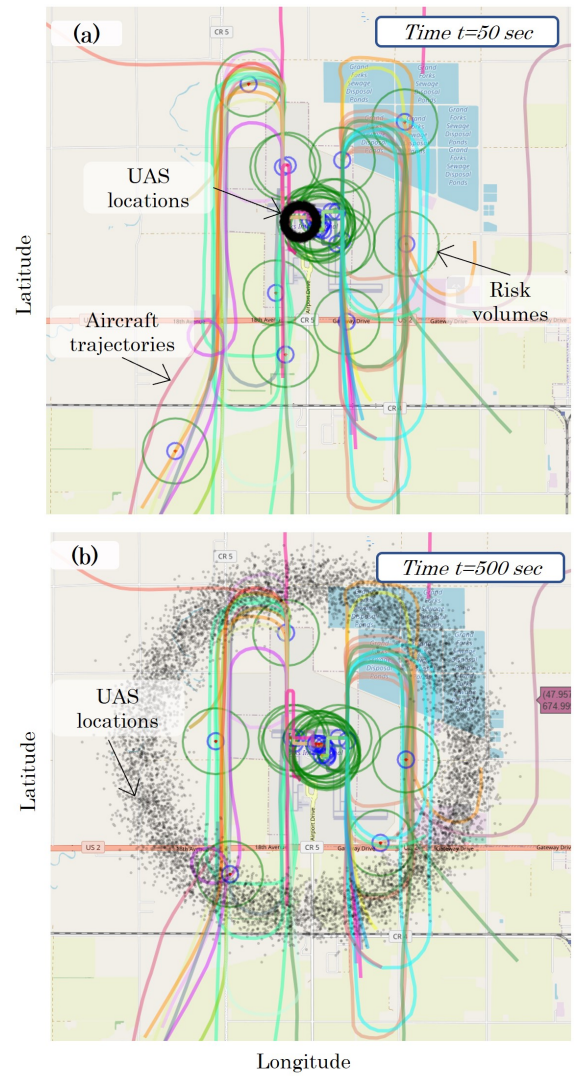


Fig. 9. The figure shows the 2D top view of simulation scenario-II; crewed aircraft paths, risk volumes, and probable UA positions (a) at time t=50 sec and (b) at time t=500 secs starting from 10:00:00 CST

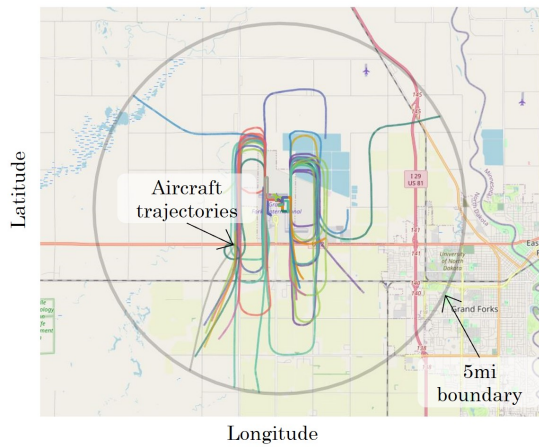


Fig. 8. Flight trajectories extracted in the given time frame. Data source: ADSB data obtained from flightradar24.com

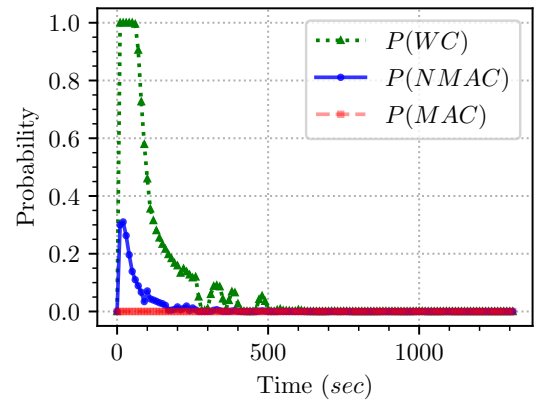


Fig. 10. Probabilities of MAC, NMAC, and WC during the peak traffic volume of the scenario-II

generated that gives uniform squares with dimensions approximately near to 0.15 mi each side, Each intersection point of the grid is taken as a failure location of UA. The simulation is executed based on the same parameters in the preceding sections.

As shown in section IV-B, the location of UAS failures plays a crucial role in determining the risk probability. The contour-map of the $P(WC)$ is shown in figure 11. The map illustrates that the maximum $P(WC)$ value is near the airport center, with the highest value of 1.0. The $P(WC)$ value gradually decreases as the failure points are away from the airport center. The $P(WC)$ value declines and becomes less than 0.0001 after approximately 3 miles. The near-miss collision $P(NMAC)$ is similarly highest at the center of the airport, with a value of 0.54, and has reduced while considering the UA failure locations away from the airport. Interestingly, there is a location *A* marked in figure 11-(b) which is far from the airport center and runways show a higher risk probability than the surrounding areas. This is the point where more incoming air traffic is descending to land towards the runway. These results highlight the importance of this research, as it is not always apparent to identify such high-risk areas far from the airport center. Nevertheless, the analysis provided here can reveal such hidden high-risk locations, which can be utilized to optimize UAS operations within airports.

D. Parameter sensitivity analysis

The findings of our analysis reveal that the mean and standard deviations associated with the heading direction and velocity variables have a substantial influence on the estimated probabilities. To further examine the impact of each of these parameters on the probabilistic risk, we have systematically varied these parameters within the context of our simulation scenario. The following parameter sensitivities are studied i) Mean heading direction ii) Mean velocity iii) Velocity standard deviation iv) Altitude mean and v) Altitude standard deviation. The specific details of the parameters studied are presented in the table III. The rest of the parameters remain the same as given in the table II.

TABLE III
PARAMETERS USED IN THE PARAMETER SENSITIVITY ANALYSIS

Parameter	Values	Units
UAS Heading mean	[0,15,30,45 ... 345]	Degrees
UAS Velocity mean	[10,20,30,40,50]	ft/sec
UAS Velocity S.D.	[2,4,6,8,10]	ft/sec
UAS Velocity S.D.	[50,100,150,200,250,300]	ft
UAS Altitude S.D	[0.5, 5,10,15,20]	Degrees

1) *Parameter sensitivity - Heading mean:* The mean heading value is varied from 0 degrees to 345 degrees with a standard deviation of 7.5 degrees. The results are presented in figure 12. From the figure, it can be seen that the loss of well-clearance probability $P(WC)$ will remain 1.0 irrespective of in which direction UA moves after the failure. The important observation is that near-mid air collision risk $P(NMAC)$ is direction dependent as shown in the figure 13. $P(NMAC)$

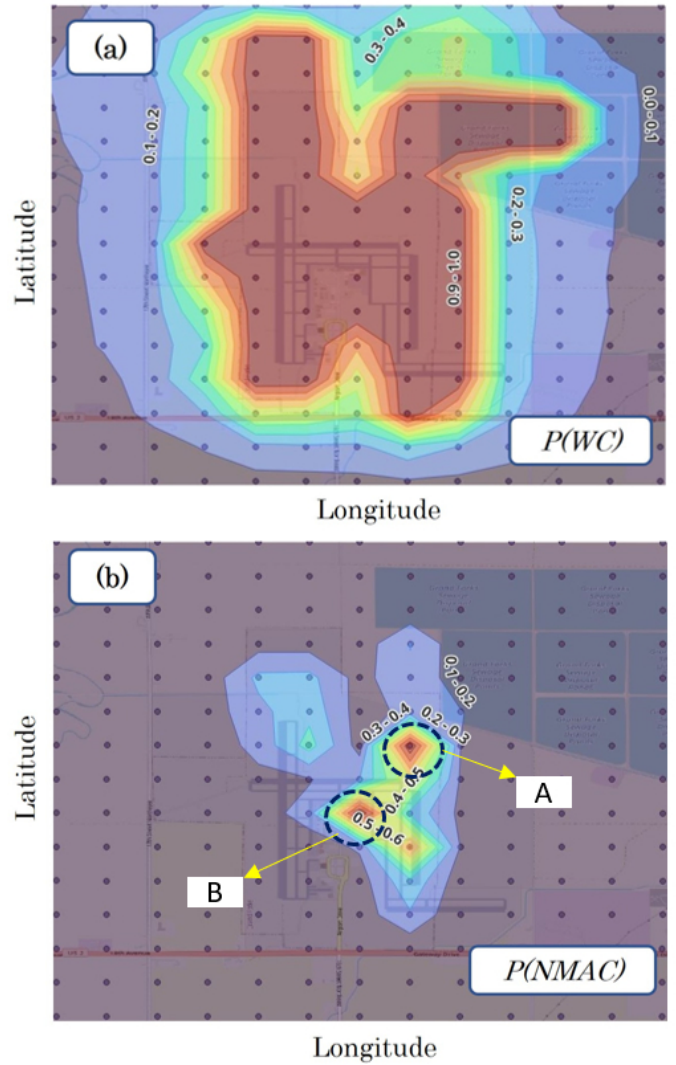


Fig. 11. Risk probability dependence on the UAS failure location simulated in scenario-III (a) risk probability associated with well-clearance (b) risk probability associated with NMAC

is high when the failed UA moves towards the east, or more specifically between north-east to south-east directions (30-135 degrees). $P(NMAC)$ is lowest when the UA navigates towards the north or south. This directional-dependent probability is highly dependent on the location of the UAS failure in relation to moving traffic. In this particular case of the current study, the majority of aircraft are located on the eastern side of the UAS failure location, which most likely is the reason for the relatively higher risk of NMAC.

2) *Parameter sensitivity - Velocity mean:* The study examines the impact of UAS failure with a change in velocity. A velocity range from 10 to 60 ft/s with an increment of 10 ft/s is considered. The associated probabilities of loss of well-clearance (WC), near mid-air collision (NMAC), and mid-air collision (MAC) are calculated. The results are shown in figure 14. The findings indicate that the probability of $P(WC)$ and $P(MAC)$ remains constant at 1.0 and 0 respectively,

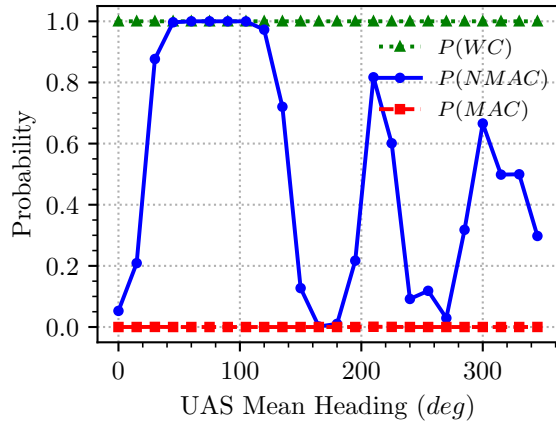


Fig. 12. Parameter sensitivity - Probability of NMAC based on the UAS mean heading direction

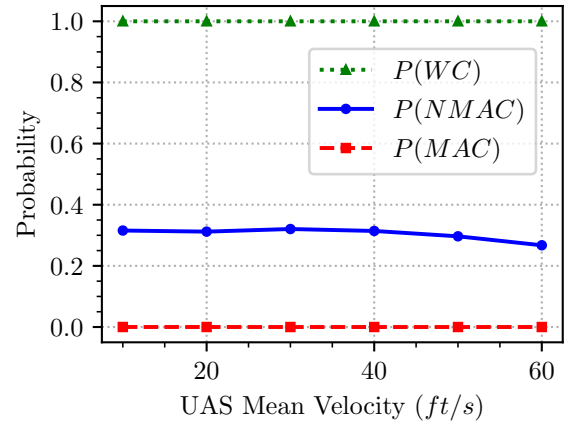


Fig. 14. Parameter sensitivity - Risk probabilities associated with the velocity of the UAS

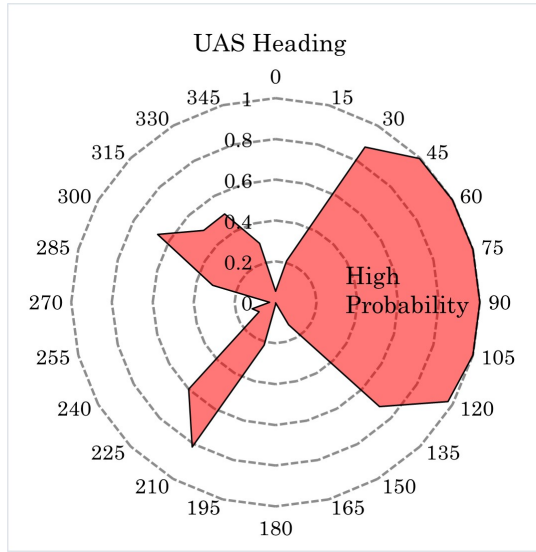


Fig. 13. Parameter sensitivity - Probability of NMAC based on the UAS mean heading direction presented in polar plot

UAS does not significantly alter the $P(WC)$ value but does influence the probability of NMAC, which declines slightly as the UAS changes its velocity after encountering a failure.

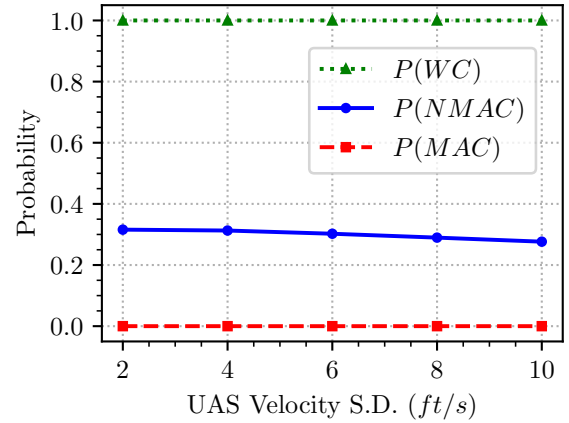


Fig. 15. Parameter sensitivity - Risk probabilities associated with the velocity standard deviation of the probability distribution

regardless of the UAS velocity. In contrast, $P(NMAC)$ decreases slightly as the mean velocity of the UAS increases.

3) *Parameter sensitivity - Velocity standard deviation:* In this case, the impact of the standard deviation of velocity normal distribution on the probabilities has been explored. Specifically, the standard deviation values are varied from 2 ft/s to 10 ft/s, while keeping the mean velocity value constant. The outcomes are depicted graphically in figure 15. Consistent with our previous findings, the well-clear probability, $P(WC)$, remains invariant at 1.0, irrespective of the velocity standard deviation. However, the value of $P(NMAC)$, decreases marginally from 0.34 to 0.26, indicating that the NMAC risk probability exhibits a slight reduction as the uncrewed aircraft alters its velocity following a failure. The results imply that altering the velocity standard deviation of

4) *Parameter sensitivity - Altitude mean:* In this scenario, the impact of the flying altitude of the mission is examined. As shown in the figure 16, $P(WC)$ and $P(NMAC)$ are independent of the altitude mean, whereas $P(NMAC)$ is dependent on the flying altitude of the UA. More specifically $P(NMAC)$ is high when the altitude is low and gradually decreases with an increase in the altitude. One possible explanation for this behavior could be the high density of moving traffic at low altitudes, as many aircraft tend to spend a relatively long time landing, taking off, and taxiing in the airport.

5) *Parameter sensitivity - Altitude standard deviation:* The influence of altitude standard deviation is investigated in this case. As shown in figure 17, $P(WC)$ and $P(NMAC)$ decrease slightly with the increase in the altitude value. This

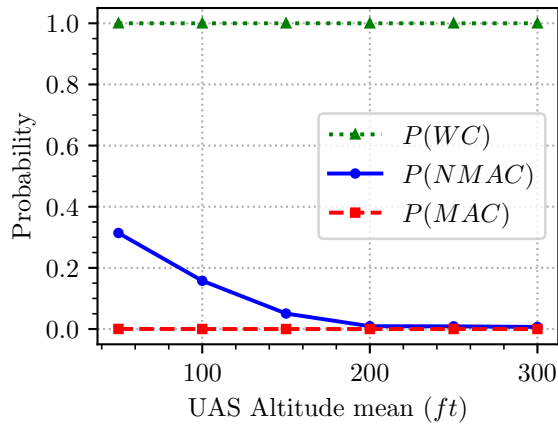


Fig. 16. Parameter sensitivity - Risk probabilities associated with the altitude mean of the probability distribution

is because of the presence of more crewed traffic at lower altitudes.

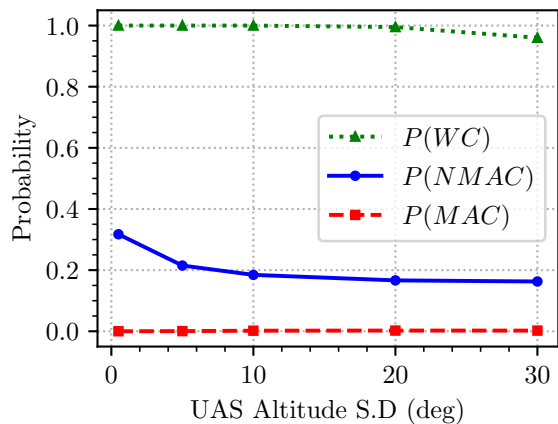


Fig. 17. Parameter sensitivity - Risk probabilities associated with the altitude standard deviation of the probability distribution

V. CONCLUSIONS

The accidental failure of UAs can cause various dangers, especially in high airport operations settings, such as risks to the infrastructure, the ground, and the air. The paper aims to create a tool for evaluating the likelihood of UAS collisions with crewed aircraft in an airport context. In order to evaluate the risks associated with uncrewed aerial systems (UASs) in mid-air collisions, near mid-air collisions, and loss of well-clearance incidents, a simulation framework has been created by incorporating real ADS-B data of crewed traffic during peak operating hours. This framework is designed to determine the probabilities of these incidents occurring, based on a stochastic analysis of the possible positions of the UAS over a given time period. The outcomes of the study show that the probabilities of risk are time-dependent. The most elevated risk transpires

when the UAS approaches the proximity of crewed aircraft. Notably, the UAS's failure time does not have any effect on the maximum probability of risk. The UAS failure location simulations indicate that the areas in close proximity to the center of the airport and runways carry the greatest risk. Moreover, the current framework has the potential to identify hidden high-risk zones that may also be considered before operating UAs. The outcomes of the parameter sensitivity analysis revealed that the mean heading direction of the UA following failure is a critical determinant of air risk. In contrast, the parameters of the probability function do not have a significant effect on the risk probabilities. Future studies may include analyzing multiple UAS operating in congested environments, such as airports, to determine saturation levels. Further research can integrate current airport capacity computations to help determine the impacts of increased UAS and AAM activity near airports.

FUNDING

This material is based upon work supported by the Federal Aviation Administration under the Alliance for System Safety of UAS through Research Excellence (ASSURE) Contract No. - A31_A11L.UAS.72. Although the data and information contained herein has been produced or processed from sources believed to be reliable, the Federal Aviation Administration makes no warranty, expressed or implied, regarding the accuracy, adequacy, completeness, legality, reliability or usefulness of any information, conclusions or recommendations provided herein. Distribution of the information contained herein does not constitute an endorsement or warranty of the data or information provided herein by the Federal Aviation Administration or the U.S. Department of Transportation. Neither the Federal Aviation Administration nor the U.S. Department of Transportation shall be held liable for any improper or incorrect use of the information contained herein and assumes no responsibility for anyone's use of the information. The Federal Aviation Administration and U.S. Department of Transportation shall not be liable for any claim for any loss, harm, or other damages arising from access to or use of data or information, including without limitation any direct, indirect, incidental, exemplary, special, or consequential damages, even if advised of the possibility of such damages. The Federal Aviation Administration shall not be liable to anyone for any decision made or action taken, or not taken, in reliance on the information contained herein.

ACKNOWLEDGMENT

The authors greatly appreciate the support provided by Sabrina Saunders-Hodge, Matthew Novak, Richard Lin, Keri Lyons, Michael DiPilato from the Federal Aviation Administration. The authors wish to acknowledge the support of their colleagues, Catherine Cahill, Tom Haritos, Katie Silas, Travis Balthazor, Henry Cathy, Jerry Hendrix, Bob Mead, Kim Kenville, Craig Carlson, James Moe, Dan Myles, Leigh Alison Jones, Hannah Thach, and Stephen 'Lux' Luxion. The authors also acknowledge SIMLAT simulation for providing

high-performance computing simulation that has contributed to the research results reported within this paper.

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